

Portfolio Code Walkthrough & Gap Analysis

From Phase 0 baseline to current device-engineer portfolio

Site	https://mirwaissarwary.github.io/
Repo	https://github.com/mirwaissarwary/mirwaissarwary.github.io
Baseline tag	phase0-quiet-medtech-baseline-2026-09-08
Baseline files	samples/phase0-baseline-2026-09-08/
Stack	HTML + CSS + vanilla JS (GitHub Pages, no framework)
Theme tokens	navy #0b1220 / teal #2ec4b6 (internal nickname: Quiet Medtech)

Tone: Tech Academy study notes -- commented steps, plain functions, no frameworks. Written so you can walk a hiring manager from the frozen Phase 0 baseline to HEAD and explain every deliberate change. The live site is a portfolio website; "Quiet Medtech" is an internal theme nickname only, not a public product label.

1. Purpose

This document is a living gap analysis: original (TTA Matrix / early redesign) -> Phase 0 Quiet Medtech baseline freeze -> current Sep 2026 portfolio. Use it two ways:

- Coding narrative: what changed in index.html, css/portfolio_style.css, js/portfolio_JS.js, and the new cases/ pages -- and why each change exists.
- Interview teaching notes: short, honest talking points for "walk me through your portfolio" without overselling or inventing frameworks.

Public materials stay engineer-first. About copy frames OptalMed as regulated-product fluency that sharpens R&D and V&V judgment -- never as a staffing or availability story.

Teaching tip: Interview opener: "It is a static GitHub Pages site. I own the HTML structure, CSS presentation, and JS behavior -- and I kept a Phase 0 baseline so I can show the before/after, not just the finished look."

2. Architecture map

Everything that ships is in the GitHub Pages root. No bundler, no React. Browser loads three primary files plus assets.

index.html -- Markup / IA: intro gate, navbar, hero, About, Projects, Coursework, Collaborate, contact form, footer buffer

css/portfolio_style.css -- Live theme: neon Matrix base + Quiet Medtech (--qm-*) overrides layered at the bottom

js/portfolio_JS.js -- Behavior: contact popup, coursework slideshow, Earth-mp4 intro Search/Skip timeline, ?portfolio=1 one-shot skip

cases/ -- Device case pages: midlabs.html, procept.html (+ cases.css); back-links use ?portfolio=1#Projects

images/ + images/introl/ -- Hero, about, collaborate art; gate poster + Earth mp4 for intro

archive/tta/ -- Frozen TTA Matrix site -- do not restyle; soft-linked from Projects

samples/phase0-baseline-2026-09-08/ -- Pre-Projects-proof freeze of index/css/js (+ stale root CSS copy)

Videos/ -- Legacy Matrix mp4 still present (known debt; not the live intro)

portfolio_style.css (repo root) -- Stale unused copy; live page links ./css/portfolio_style.css

Teaching tip: When editing, always open css/portfolio_style.css under the css/ folder. The root portfolio_style.css is a leftover and is not what the live page loads.

3. Page load / intro gate (current)

On a normal visit (no ?portfolio=1), the body starts with class Intro_Active and Portfolio_Main is hidden. Deferred JS wires Search and Skip after DOMContentLoaded.

Current Search timeline

- Gate: Earth mp4 (Intro_Gate_Video) is shown PAUSED on the first frame behind real Search / Skip controls.
- Search: video plays once; Intro_Status flashes "Searching..."
- Found: after ended (or safety timer ~12.5s), status shows "Candidate found" then the name WITH the Earth video still visible but paused behind the text -- not a hard cut to a JPG-only stage.
- Frame 3: short handoff into the portfolio hero (medical-equipment background + pills).

- Skip: instant showPortfolio() -- no animation obligation.

Case back-link one-shot skip

- MID Labs / PROCEPT "Back to Projects" links go to index.html?portfolio=1#Projects.
- JS cameFromCaseBackLink() detects portfolio=1, calls skipIntro(), then history.replaceState clears the query so a later refresh returns to the gate.
- Plain refresh or a bare #hash always shows the gate again -- skip is one-shot from case pages only.

Teaching tip: prefers-reduced-motion skips the video play-through and uses a short Candidate found -> name -> portfolio path.

4. Gap analysis table

Compare three eras: TTA Matrix / early redesign, the Phase 0 baseline freeze (tag phase0-quiet-medtech-baseline-2026-09-08), and HEAD as of Sep 2026.

Topic	TTA / early	Phase 0 baseline	Current (Sep 2026)
Visual system	Neon Matrix green / magenta chrome	Quiet Medtech navy/teal overrides on top of neon base	Same dual stack; teal form buttons added; public label is portfolio site (not product name)
Intro	Matrix splash / early JPG stage gate	Earth video gate; Search/Skip; JPG stage swaps	Earth mp4 paused at gate; plays on Search; paused video stays behind found/name; Skip instant; ?portfolio=1 one-shot
Projects	TTA live projects featured	Same TTA-forward cards + archive mention	MID Labs + PROCEPT device cases lead; Device Telemetry Lab, SecureTodoApi, site; TTA demoted to soft links
Coursework carousel	TTA certificate Skills slideshow	Same Skills carousel as primary skills story	Carousel kept/restored as "Software developer coursework" -- not a Skills grid; broader skills live in About + cases
Identity copy	Mixed student / Matrix identity	Engineer title; About leaned on bold RA/QA Manager hook	Engineer who codes first; OptalMed without bold RA/QA hook; pills: Safety and Quality + Systems & Software
Form buttons	Neon green SUBMIT / magenta CLOSE	Same neon/magenta (override not yet applied)	Teal SUBMIT + navy/teal CLOSE via Quiet Medtech overrides at end of CSS
Section order	TTA-era Skills-heavy flow	About -> Skills -> Projects-ish ordering	About -> Projects -> Coursework -> Collaborate; nav matches (Dev coursework)
Case pages	None (TTA project cards only)	None	cases/midlabs.html + cases/procept.html with interview-ready R&D narrative + back-link skip
Footer buffer	Busy social / title footer	Blank footer-buffer after cleanup	Kept as quiet frame (aria-hidden); sticky nav covers navigation

5. Change log by phase (0-5)

Phases below match the work landed after the Phase 0 tag through HEAD. Dates are Sep 2026 local work; commit subjects are paraphrased for teaching.

Phase 0 -- Baseline freeze

- Tagged phase0-quiet-medtech-baseline-2026-09-08.
- Copied index.html, css/portfolio_style.css, js/portfolio_JS.js, and root stale CSS into samples/phase0-baseline-2026-09-08/ for before/after PDF and interview demos.
- Purpose: never lose the "Quiet Medtech but pre-proof" snapshot while Projects evolved.

Phase 1 -- Projects proof (device + code)

- First featured software proof: Device Telemetry Lab, SecureTodoApi, and this portfolio site (plus GitHub profile card). TTA live projects / Matrix archive demoted to soft links under Projects.
- Then device cases took the lead: MID Labs (MVE / UVE / CIS) and PROCEPT AquaBeam R&D case pages under cases/, linked as featured project cards.
- Projects intro copy: interview-walkable device R&D + software samples -- truthful, not a fake product catalog.

Phase 2 -- Coursework carousel (not Skills grid)

- A Quiet Medtech "skills grid" was tried, then dropped -- the TTA certificate carousel was restored because it is honest coursework evidence, not a fake competency matrix.

- Relabeled and scoped as "Software developer coursework" (nav: Dev coursework). Broader engineering skills are told in About + device cases, not badge spam.

Phase 3 -- Identity copy

- Hero + About lead with biomedical / medical-device engineer who also codes.
- OptalMed mentioned as regulated-product / quality-system fluency without a bold RA/QA Manager hook.
- Hero pills include Safety and Quality and Systems & Software (alongside User Centered Design, Engineering Excellence, Concept to Market).
- Dropped "Quiet Medtech" as a public portfolio product label; site reads as a personal portfolio website.

Phase 4 -- Visual system (teal form chrome)

- Contact form SUBMIT / CLOSE restyled with teal / navy Quiet Medtech overrides (CSS comment block: Phase 4) so chrome matches the rest of the adult theme.
- Neon green + magenta rules still exist higher in the file -- overrides win by cascade. That dual personality is an explicit known debt (see section 7).
- Section order aligned: About -> Projects -> Coursework -> Collaborate.

Phase 5 -- Mobile polish + intro / back-link behavior

- Mobile: tighten nav wrap; hide duplicate Contact nav item and floating CONTACT button on small screens.
- Intro: keep paused Earth video visible behind Candidate found and name (not only at the idle gate).
- Case back-links: ?portfolio=1 skips intro once into #Projects; replaceState clears the flag so refresh returns to the gate.

6. How to extend

Add a device case page

- Copy cases/midlabs.html (or procept.html) -> cases/<slug>.html; keep cases.css link.
- Write interview-walkable R&D narrative (problem, constraints, your role, outcome) -- no confidential data.
- Back link must be ../index.html?portfolio=1#Projects so return skips the gate once.
- Add a featured .project-card in index.html Projects grid pointing at the new case.

Add a software project card

- Copy an existing .project-card--featured block; point href at the public GitHub repo.
- Keep the intro sentence truthful: what it demonstrates (API, telemetry, tooling) in one breath.
- Prefer repos that already exist and build -- do not invent private work as live demos.

Republish

- Commit on main (GitHub Pages serves the repo root). Wait for Pages deploy.
- Hard-refresh the live site; if images stick, bump ?v= cache-busters on changed assets.
- Re-run this script to refresh the living PDF after meaningful IA/theme changes.

Teaching tip: Extension rule of thumb: device cases lead the story; coding projects prove you can ship software next to hardware; coursework stays scoped as coursework.

7. Known debts

- Dual CSS personality: neon Matrix base rules still sit under Quiet Medtech (--qm-*) overrides. Works, but a future cleanup pass should delete dead neon once nothing depends on it.
- Stale root portfolio_style.css: live page uses ./css/portfolio_style.css only. Root file confuses editors and should eventually be removed or clearly marked unused.
- Matrix mp4 still under Videos/: leftover from the archive era; live intro uses images/intro Earth mp4.
- More coding projects later: Telemetry Lab + SecureTodoApi are a start; add only when the repo is demoable and the card copy stays honest.
- Form markup labels still say SUBMIT / CLOSE (uppercase TTA habit); visual chrome is teal, but copy could soften in a later polish pass.

End of living walkthrough. Baseline tag: phase0-quiet-medtech-baseline-2026-09-08. Regenerate via `/workspace/portfolio-docs/build_portfolio_gap_pdf.py` with `/workspace/pdf-venv/bin/python`.